

Inference-Only Evaluation of Time-Series Foundation Models

Task, controls, and population-aware metrics

Thesis project

Standalone method note

Abstract

This note isolates the scientific formulation from implementation and cluster operations. It is intended to be readable on its own and reusable when drafting a paper introduction or method section.

1 Evaluation task

For user u and deterministic query date t ,

$$X_{u,t} = x_u(t - L + 1:t), \quad Y_{u,t} = x_u(t + 1:t + H).$$

A frozen model F_θ produces $\hat{Y}_{u,t} = F_\theta(X_{u,t}, C_{u,t})$. The evaluator controls accessible dates and covariates but does not change the model’s native inference algorithm.

2 Preprocessing and covariates

Optional target instance normalization uses $Z = (X - \mu_X)/(\sigma_X + \varepsilon)$ and reverses the transform on the forecast. Metrics are always computed on raw-scale predictions. Covariates remain on their source scale. Modes are none, target identity, or prepared known covariates; an incapable adapter rejects non-empty covariates.

3 Reference forecasts

Chronos-2 is compared with persistence, the lookback mean, repetition of the last H observations, and weekly aligned lookback persistence. A separate family compares official Chronos-2, Chronos-Bolt, Chronos-T5, and TS-ICL pipelines.

4 Metrics

For each of MSE, MAE, nMSE, nMAE, and MASE, reports include the sample mean and population dispersion, equal-user mean, dispersion across user means, worst-10-percent-user mean, and horizon-wise behavior. Timing is amortized per user and per user-window without including model or data loading.

5 Controlled questions

The four workflow families test univariate model quality, target normalization/constant-window controls, covariate use, and cross-foundation comparison. Cadence-aware short, mid, and long ranges make horizons comparable across hourly, daily, and 15-minute data.

6 Evidence boundary

Run identity contains only declared scientific configuration and seed. Publishable reports contain aggregate results and provenance; detailed per-configuration plots and aligned inputs live in the diagnostics hierarchy.